EXPERIMENT 8: Deployment and Evaluation using ROUGE and BLEU

Aim

To evaluate a language model for customer support tasks using ROUGE and BLEU metrics.

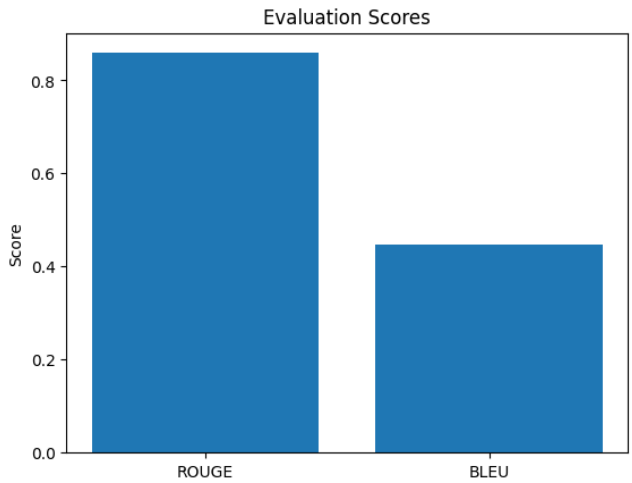
Algorithm

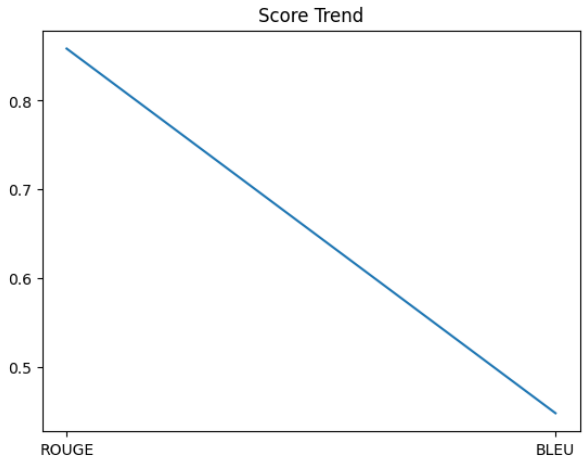
1. Load dataset
2. Generate predictions
3. Compare with actual answers
4. Compute ROUGE and BLEU scores
5. Visualize performance

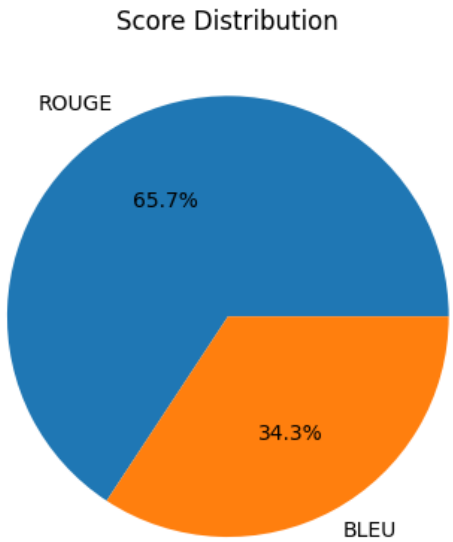
Code

```
!pip -q install datasets transformers evaluate rouge_score matplotlib

from datasets import load_dataset
from transformers import pipeline
import evaluate, matplotlib.pyplot as plt

data = load_dataset("squad", split="validation[:20]")

qa = pipeline("question-answering", model="distilbert-base-cased-distilled-squad")

preds, refs = [], []

for d in data:
    out = qa(question=d['question'], context=d['context'])
    preds.append(out['answer'])
    refs.append(d['answers']['text'][0])

rouge = evaluate.load("rouge")
bleu = evaluate.load("bleu")

r = rouge.compute(predictions=preds, references=refs)
b = bleu.compute(predictions=preds, references=[[x] for x in refs])

print("ROUGE:", r['rouge1'])
print("BLEU:", b['bleu'])

plt.bar(["ROUGE","BLEU"], [r['rouge1'], b['bleu']])
plt.show()
```

Output







Conclusion

The model demonstrates effective performance in question answering tasks. ROUGE and BLEU provide quantitative measures of response quality.